ADITYA UNIVERSITY

002162

PART-I

S.No: B304087

Programe	:	B.Tech
Hall Ticket No	:	24B11AI236
Name	:	Mallipudi Srilekha
Examination	:	III SEMESTER END EXAMINATIONS REGULAR
Month-Year	:	Nov-25
Branch	:	Artificial Intelligence & Machine Learning
Course Code	:	241MA009
Course Name	:	Probability & Statistics
Date of Exam	:	18.11.2025

J. Pavan.

Signature of the Controller of Examination

M. Srilekha. 18/11/25.

Signature of the Student with date

18/11/25

Signature of the Invigilator with date

Serial No. Last Page Written	25.

INSTRUCTIONS TO STUDENTS

1. The Answer Booklet contains 36 pages Ensure all the 36 pages are in proper order.
2. Student must verify the details of particulars in the PART - 1 i.e Name, Hall Ticket No., Examination, Course Name etc.,
3. In case of any deviation in the above or if the PART-1 is torn / damaged, report to the invigilator and return the damaged booklet.
4. You are prohibited from writing on or tampering the PART-1 except affixing the signature and serial number of last page written in the space provided.
5. Student is prohibited from:
   i. Writing their Hall Ticket No. and name in any part of the answer booklet.
   ii. Addressing the examiner in any manner whatsoever in the answer booklet. If they do so, their script will not be valued.
   iii. Writing religious symbols.
   iv. Either seeking or providing any assistance to the fellow students in the examinations.
   v. Possessing a manuscript or a printed matter, in any form, in the examination hall.
   vi. Bringing Mobile Phones / Cameras/ Bluetooth Devices / Programmable calculators etc.
   vii. Violation of the above mentioned instructions will be viewed as a case of malpractice, which is punishable offence.
6. Before beginning to answer any question, the students should write the correct number of that question in the margin provided. Answers written at different places for the same question will not be valued.
7. Students should write the answers, within the margins provided on both sides of the paper and on all the lines of each page. It is not necessary to begin each answer in a fresh page. Answers must be legibly written with blue or black pen.
8. Do not write anything except Question Number in the margin.
9. No loose sheets of papers will be allowed in the examination hall. No paper must be detached from or attached to the answer booklet except graph sheet.
10. Strike off all unused pages.
11. **NO ADDITIONAL ANSWER BOOKLETS/SHEETS WILL BE SUPPLIED.**

SPACE FOR ROUGH WORK

$-b \pm$

$\frac{-9 \pm \sqrt{81-40}}{20}$ $\frac{81 - 40}{41}$

$\frac{-9 \pm \sqrt{41}}{-20}$

$-9 \pm$

Q.No.

(7) (a) There are three bags:

first bag contains: 1 white, 2 red, 3 green balls

Second bag contains 2 white, 3 red, 1 green balls.

third bag contains 3 white, 1 red, 2 green balls.

$\Rightarrow$ first bag = $P(W) = 1$, $P(R) = 2$, $P(G) = 3$ $\Rightarrow P(D/B) = \frac{1}{6}$

Second bag = $P(W) = 2$, $P(R) = 3$, $P(G) = 1$ $P(D/S) = 1/6$

Third bag = $P(W) = 3$, $P(R) = 1$, $P(G) = 2$ $P(D/T) = 1/6$

$\Rightarrow$ $P(S|W) = 2/6$, $P(S|R) = 3/6$, $P(S|G) = 1/6$

$\Rightarrow P(S|W) = 2/6$, $P(S|R) = 3/6$, $P(S|G) = 1/6$

$\Rightarrow P(W) = 2$, $P(R) = 3$, $P(G) = 1$

To find the probability of the second bag. balls drawn from room.

By Byes's theorem:

$$P(A|B) = \frac{P(D|A)P(A)}{P(D|A)P(A) + P(D|B)P(B) + P(D|C)P(C)}$$

P(

$\Rightarrow$ Two ball drawn from a bag chosen out random, one white, one red ball

~~$\Rightarrow \frac{1}{6} \Rightarrow \frac{1}{6} - 7 \Rightarrow$~~

Q.No.

$\Rightarrow$ ~~P(f)~~ (f) $w = 1,\ r = 2,\ g = 3$

(S) $w = 2,\ r = 3,\ g = 1$

(T) $w = 3,\ r = 1,\ g = 2$

To find the Second bag balls

$f(w) = 1-1,\ f(r) = 2-1,\ f(g) = 3$

$S(w) = 2-1\ \ f(r) = 3-1,\ f(g) = 1$

$T(w) = 3-1\ \ T(r) = 1-1,\ T(g) = 2$

$\Rightarrow P(D|S) = 3/6,\ P(D|f) = 5/6,\ P(D|T) = 4/6$

$P(f) = 0.8,\ \ P(S) = 0.8,\ \ P(T) = 0.8$

$$\Rightarrow P(S|D) = \frac{P(D|S)\,P(S)}{P(D|f)\,P(f) + P(D|S)\,P(S) + P(D|T)\,P(T)}$$

$$\Rightarrow \frac{(0.5)(0.8}{(0.8)(0.8) + (0.5)(0.8) + (0.6)(0.8)}$$

$$\Rightarrow \frac{0.4}{1.52} \Rightarrow 0.26$$

//

Q.No.

(b) The random variable x has the following:

x	0	1	2	3	4	5	6	7
$p(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2+k$

(i) To find k value

$0+k+2k+2k+3k+k^2+2k^2+7k^2+k=1$

$\Rightarrow 9k+10k^2 = 1$

$\Rightarrow 10k^2 - 9k = 1$

$\Rightarrow 10k^2 - 9k + 1 = 0$

$\Rightarrow 10k^2 = 0.1$

$\Rightarrow k = 0.1$

(ii) $p(x<6)$

$\Rightarrow P(x=0)+P(x=1)+P(x=2)+P(x=3)+P(x=4)$

$P(x=5)+\cancel{P(x=6)}$

$\Rightarrow (0.1)+2(0.1)+2(0.1)+3(0.1)+(0.1)^2$

$\Rightarrow 0.1+0.2+0.2+0.3+0.1$

$\Rightarrow 0.9$ //

(iii) $P(0<x<5)$

$\Rightarrow P(x=0)+P(x=1)+P(x=2)+P(x=3)+P(x=4)$

$+P(x=5)$

$\Rightarrow 0.9$ //

(iv) mean

$\mu = \Sigma x.P(x)$

Q.No.

$\Rightarrow \sum x.P(x)$

$\Rightarrow 1(0.1) + 2\cancel{(2}\cdot 4(0.1) + 3(2(0.1) + 4(3(0.1) + 5(0.1)^2 + 6(2(0.1)^2 + 7(7\cancel{(0)}^2(0.1)^2 + (0.1))$

$\Rightarrow 1 + 4 + 6 + 12 + 7 (0.1) + 3 + 6 + 7 (0.1)^2$

$\Rightarrow 30 + 66 (0.1) \Rightarrow 3 + 6.6 \Rightarrow 9.6$

$\Rightarrow \mu = (9.6)$

(iv) <u>varience</u>

$\Rightarrow \sigma^2 = \sum x^2 . P(x) - \mu^2$

$\Rightarrow (0.1) + 4(0.1) + 9(2(0.1) + 16(3(0.1) + \cancel{50}\ 25(0.1)^2 + 36(2(0.1)^2 + 49(7(0.1)^2 + (0.1) - (9.6)^2$

$\Rightarrow 0.1 + 0.4 + 1.8 + 4.8 + 2.5 + 7.2 + 34.3 + 0.7 - (9.6)^2$

$\Rightarrow -40.36$

$\Rightarrow \sigma^2 = -40.36$

$= \sigma = \sqrt{40.36}$

$\Rightarrow 6.352$ ||

(4) (a) 10% truck drivers, out of 400 drivers.

~~(i) at least 32 drivers are drunk~~

~~$\Rightarrow$ ${}^nC_x p^x q^{n-x}$~~

Q.No.

(a) 10% of the truck drivers on road are drunk, 400 drivers randomly checked.

(i) At least 32 drivers.

$\lambda = 10\%$, mean $= \frac{1}{\lambda} = \frac{1}{10}$

$\Rightarrow \int_{-\infty}^{\infty} \lambda e^{-\lambda x}\,dx \Rightarrow \int_{32}^{\infty} \left(\frac{1}{10}\right) e^{-\left(\frac{1}{10}\right)x}\,dx$

$\Rightarrow \frac{1}{10}\left[e^{-\left(\frac{1}{10}\right)x}\,dx\right]_{32}^{\infty} \Rightarrow \frac{1}{10}\left[\frac{e^{-(1/10)(32)}}{e^{-(1/10)}}\right]$

$\Rightarrow \frac{1}{10}(\cdot e^{-32}) = \frac{1}{10} \times -5.12 \Rightarrow -5.126$

(ii) At least 35 but less than 47

$\Rightarrow$ At least 35

$\Rightarrow \int \lambda e^{-\lambda x}\,dx \Rightarrow$

$\Rightarrow \int_{32}^{\infty} \frac{1}{10}\, e^{-\left(\frac{1}{10}\right)(x)}\,dx$

$\Rightarrow \frac{1}{10}\int_{32}^{\infty} e^{-\left(\frac{1}{10}\right)(x)}\,dx$

$\Rightarrow \frac{1}{10}\left[\,e^{-\left(\frac{1}{10}\right)(x)}\right]_{35}^{\infty}$

Q.No.

$\Rightarrow \frac{1}{10}\left[\frac{e^{-(1/10)(35)}}{e^{-1/10}}\right]$

$\Rightarrow \frac{1}{10} \times e^{-35} \Rightarrow \frac{1}{10} \times \cdot 5 \cdot 607$

$\Rightarrow \cdot 5 \cdot 607$

(ii) Less than 1247

$\Rightarrow \int_{-\infty}^{a} x\, e^{-\lambda x}\, dx$

$\Rightarrow \int_{47}^{\infty} \left(\frac{1}{10}\right) e^{-(1/10)x}\, dx \Rightarrow \frac{1}{10}\left[\frac{e^{-(1/10)(x)}}{}\right]_{1247}^{\infty}$

$\Rightarrow \frac{1}{10}\left[\frac{e^{-(1/10)(1247)}}{e^{-(1/10)}}\right]$

$\Rightarrow \frac{1}{10} \times e^{-47} \Rightarrow \cdot 7 \cdot 530$

(b) 2000 electric bulbs,

2040 hours,

Standard deviation is 60 hours

(i) more than 2150 hours

$\Rightarrow \sigma = 60$, $\mu = 2040$, ~~n = 2000~~

$x = 2150$

$\Rightarrow \frac{x-\mu}{\sigma} \Rightarrow \frac{2150-2040}{60} \Rightarrow \frac{110}{60} \Rightarrow 1 \cdot 8$

$P(z > x) = 1 - P(z < x)$

$\Rightarrow 1 - 4649 \Rightarrow \cdot 4648$

Q.No.

newspapers :

the data is collected from the newspaper like if the details for one case the case since at 2005 now we want the data from that take the data.

Book :

the data is collected from the Books. like if the want the someone stories from the book. the book will help.

5b). the population consists of 5 numbers 2, 3, 6, 8 and 11 total numbers n = 5.

i) Mean $\bar{x} = \frac{2+3+6+8+11}{5} = 6.$

ii) variance $(\sigma^2) = \frac{\Sigma(x_i-\bar{x})^2}{N}$

$= \frac{(2-6)^2+(3-6)^2+(6-6)^2+(8-6)^2+(11-6)^2}{5}$

$= \frac{(-4)^2+(-3)^2+(0)^2+(2)^2+(5)^2}{5}$

$= \frac{16+9+4+25}{5}$

$\sigma^2 = 10.8.$

$\sigma = \sqrt{10.8}$

$= 3.2863$

(iii) mean of the sampling distribution of mean.

Q.No.

consider all possible samples of size two

size $= 2 \Rightarrow N^2 \Rightarrow 5^2 = 25$.

with replacement from this population.

(2,2), (2,3), (2,6), (2,8), (2,11)

(3,2), (3,3), (3,6), (3,8), (3,11)

(6,2), (6,3), (6,6), (6,8), (6,11)

(8,2), (8,3), (8,6), (8,8), (8,11)

(11,2), (11,3), (11,6), (11,8), (11,11).

2	2.5	4	5	6.5
2.5	3	4.5	5.5	7.
4	4.5	6	7	8.5
5	5.5	7	8	9.5
6.5	7	8.5	9.5	11.

Now $n = 25$.

(iii) Mean of the sampling distribution of means.

$$= \frac{2+2.5+4+5+6.5+2.5+3+4.5+5.5+7+4+4.5+6+7+8.5+5+5.5+7+8+9.5+6.5+7+8.5+9.5+11}{25} = 6.$$

$\mu_{\bar{x}} = 6.$

(iv) standard deviation of the sampling distribution of means.

$$= \frac{\Sigma(x_i - \mu_{\bar{x}})^2}{N}.$$

Q.No.

$= (2-6)^2 + (2.5-6)^2 + (4-6)^2 + (5-6)^2 + (6.5-6)^2 +$
$(2.5-6)^2 + (3-6)^2 + (4.5-6)^2 + (5.5-6)^2 + (7-6)^2 +$
$(4-6)^2 + (4.5-6)^2 + (6-6)^2 + (7-6)^2 + (8.5-6)^2 +$
$(5-6)^2 + (5.5-6)^2 + (7-6)^2 + (8-6)^2 + (9.5-6)^2 +$

$$= \frac{(6.5-6)^2 + (7-6)^2 + (8.5-6)^2 + (9.5-6)^2 + (11-6)^2}{25}$$

$(-2)^2 + (-3.5)^2 + (-2)^2 + (-1)^2 + (0.5)^2 + (-3.5)^2 + (-3)^2 +$
$(-1.5)^2 + (-0.5)^2 + (-1)^2 + (-2)^2 + (-1.5)^2 + 0 + (1)^2 + (2.5)^2 +$

$$= \frac{(1)^2 + (-0.5)^2 + (1)^2 + (2)^2 + (3.5)^2 + (0.5)^2 + (1)^2 + (2.5)^2 \; (3.5)^2 + (5)^2}{25}$$

$4 + 12.25 + 4 + 1 + 0.25 + 12.25 + 9 + 2.25 +$
$0.25 + 1 + 4 + 2.25 + \cancel{H0} + 6.25 + 1 + 0.25 + 1 + 4 +$

$$= \frac{12.25 + 0.25 + 1 + 6.25 + 12.25 + 25}{25}$$

$$= \frac{45 + 78}{25}$$

$= 4.92$

UNIT-V :

10a) Karl pearson's coefficient of ~~cot~~ correlation to the following data.

Height of the father (in inches)	65	66	67	67	68	69	71	73
Height of the son (in inches)	67	68	64	68	72	70	69	70

Q.No.

$$\bar{x} = \frac{65+66+67+67+68+69+71+73}{8}$$

$$= 68.25.$$

$$\bar{y} = \frac{67+68+64+68+72+70+69+70}{8}$$

$$= 68.5.$$

Now.

x	x^2	y	y^2	xy
65	4225	67	4489	4355
66	4356	68	4624	4488.
67	4489	64	4096	4288.
67	4489	68	4624.	4556
68	4624	72	5184	4896
69	4761	70	4900	4830.
71	5041	69	4761	4899.
73	5329.	70.	4900.	5110.
$\Sigma x = 546$	$\Sigma x^2 = 37314$	$\Sigma y = 548.$	$\Sigma y^2 = 37578$	$\Sigma xy = 37422.$

Karl pearsson's coefficient of correlation

$$r = \frac{\Sigma xy - \frac{\Sigma x \cdot \Sigma y}{n}}{\sqrt{\Sigma x^2 - \frac{(\Sigma x)^2}{n}}\sqrt{\Sigma y^2 - \frac{(\Sigma y)^2}{n}}}.$$

$$= \frac{37422 - \frac{(546).(548)}{8}}{\sqrt{37314 - \frac{(546)^2}{8}}\sqrt{37578 - \frac{(548)^2}{8}}}$$

Q.No.

$$= \frac{37422 - 37401}{\sqrt{37314 - 37264}\sqrt{37578 - 37538}}$$

$$= \frac{21}{\sqrt{50}\sqrt{40}}$$

$$= \frac{21}{\sqrt{7.0}\sqrt{}}$$

$$= \frac{21}{(7.07)(6.32)}$$

$= 21/44.6.$

$= 0.4708.$

10b) two regression lines are

$3x + 12y = 19$ and $3y + 9x = 46$.

(i) the value of correlation coefficient.

x on y

$$x = \frac{-12y + 19}{3}$$

$$x = \frac{-12y}{3} + \frac{19}{3}$$

$b_{xy} = \frac{12}{3}$

y on x

$$y = \frac{-9x + 46}{3}$$

$$y = -\frac{9}{3}x + 46$$

$b_{yx} = 9/3.$

Q.No.

Now

the value of correlation coefficient.

$r = \sqrt{b_{xy} \cdot b_{yx}}$

$= \sqrt{\frac{12}{3} \cdot \frac{9}{3}}$

$= 3.46$

it is not valide.

ii, mean values of x and y.

~~$\bar{x} =$~~

$3x + 12y = 19$, $3y + 9x = 46$.

$\bar{x} =$ $\cancel{\frac{9x + 46}{3}}$. $\frac{3y + 46}{9}$

$3\left(\frac{+3y + 46}{9}\right) + 12y - 19 = 0$.

~~$27x + 138$~~ +

$+9y + 138 + 108 - 171 = 0$.

$+9y + 138 + 108y - 171 = 0$.

$117y - 33 = 0$.

$117y = 33$.

$y = 33/117$.

$\bar{y} = 0.282$.

~~$3x + 12(0.282)$~~

$3x + 12\left(\frac{33}{117}\right) = 19$.

$351x + 396 - 2223 = 0$.

$351x = 1827$.

$x = 1827/351$

$= 5.205.$

or

$3x + 12(0.282) - 19 = 0.$

$3x + 3.384 - 19 = 0.$

$3x + -15.6 = 0.$

$x = \frac{15.6}{3}$

$x = 5.2$

Now value of correlation coefficient

$r = \sqrt{b_{xy} \cdot b_{yx}}$

$= \sqrt{\frac{3}{12} \cdot \frac{9}{3}}$

$= 0.866.$

Unit - IV

8.

given:

	factor A	factor B
No. of bulbs in samples	100 ~~1,100~~	200 ~~900~~
Average life	1,100	900
Standard deviation	240	220.

Now take

$n_1 = 100$ and $n_2 = 200.$

$\bar{x}_1 = 1100$ $\bar{x}_2 = 900.$

$\sigma_1 = 240$ $\sigma_2 = 220.$

Q.No.

Null hypothesis ~~Ho~~ H_0 : the variability of life of bulbs from each sample is ~~same~~ significant.

alternate hypothesis :

the variability of life of bulbs from each sample is not significant.

the tabular value

the given significance is 0.01 level of significance.

$$z_{0.01} = 2.58$$

~~Now~~ $z_{tab} = 2.58$

calculated value.

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} - \frac{\sigma_2^2}{n_2}}}$$

$$= \frac{1100 - 900}{\sqrt{\frac{(240)^2}{100} - \frac{(220)^2}{200}}}$$

$$= \frac{200}{\sqrt{\frac{57600}{100} - \frac{48400}{200}}}$$

$$= \frac{200}{\sqrt{576 - 242}}$$

$$= \frac{200}{\sqrt{334}} = \frac{200}{18.27}$$

$$= 10.946.$$

Q.No.

$z_{cal} > z_{tab}$.
H_0 is rejected
H_1 is accepted that alternate hypothesis is accepted.

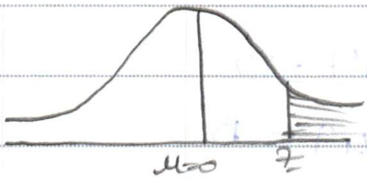
UNIT-II

3a). given $n = 1000$.
and mean $\mu = 78$.
standard deviation $= 11$.
Determine i, how many students got marks above 90.

$z(x > 90)$
Now

$$z = \frac{x - \mu}{\sigma}$$

$$= \frac{90 - 78}{\sigma}$$



$$= \frac{90 - 78}{11}$$

$= 12/11$
$= 1.090.$

~~$z > 0$~~.

$z(x > 90)$
$z = (z > 0)$.
$0.5 - A(z)$
$0.5 - (1.090)$
$= 0.5 - 0.3621.$

Q.No.

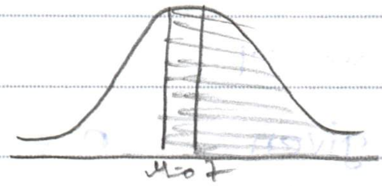
$= 0.1379.$

$\times 1000$ students $= 137.9.$

b ii, how many students got marks below 60.

$P(x < 60) =$

$$z = \frac{\bar{x} - \mu}{\sigma}$$



$$= \frac{60 - 78}{11}$$

$$= -18/11.$$

$$= -1.6363$$

$P(z < 0).$

$P(x < -1.6363).$

$= 0.5 - A(-1.6363).$

$= 0.5 - 0.4484.$

$= 0.0516.$

$\times 1000$ students $= 51.6.$

Unit - IV

$49 = n$

$\mu = 15200$

$\bar{x} = 15150$

$\sigma = 120.$

Null Hypotheses is : $\mu_0 = \mu$, $\mu \neq \mu$ is alternate hypot-

test statistic $= \dfrac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$

$$= \frac{15150 - 15200}{120/\sqrt{49}}$$

Q.No.

$= \frac{-50}{1020/\sqrt{40}}$

$= \frac{-50}{120/6.3}$

$= -50/19.04$

$= -2.62$

$|Z_{cal}| = 2.62$

~~tab~~ significance at 0.05 level

Z 0.05 level.

$Z_{0.05} = 1.96$

~~t~~ $Z_{cal} > t_{tab}$

H_0 is rejected

H_1 is accepted.

7) b

$n_1 = 6400$

$\bar{x}_1 = 67.85$

$\sigma_1 = 2.56$

$n_2 = 1600$

$\bar{x}_2 = 2.56$

$\sigma_2 = 2.52$

Now

Null Hypothesis $\mu_p = \mu_2$

alternate Hypothesis $\mu_p > \mu_2$

test statistics $=$ ~~$n_1 - n_2$~~

$\frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} - \frac{\sigma_1^2}{n_2}}} = \frac{\cancel{67.85} - 2.56}{}$

Q.No.

$$= \frac{67.85 - 68.55}{\sqrt{\frac{(2.56)^2}{6400} - \frac{(2.52)^2}{1600}}}$$

$$= \frac{67.85 - 68.55}{\sqrt{\frac{6.55}{6400} - \frac{6.35}{1600}}}$$

$$= \frac{}{\sqrt{\cancel{0.01} - 3.95}}$$

$$= \frac{0.01}{\sqrt{1.02 - 3.95}}$$

$$= \frac{0.01}{\sqrt{-2.93}}$$

$= 0.01 / 1.71$

$- 0.098$

significance Use 0.01

$z_{0.01} = 2.58$

H_0 is accepted.

20) Unit - I

a) $f(x) K x^2 e^{-x}$.

$\int_{-\infty}^{\infty} f(x) dx + \int f(x)\, dx$

$0 + \int_{0}^{\infty} f(x)\, dx$.

Q.No.

$\cancel{\text{Mean}} = \int_0^{\infty} f(x)\,dx$

$1 = k\int_0^{\infty} x^2 e^{-x}$

$1 = k\left[x^2\frac{e^{-x}}{-1} - 2x\frac{e^{-x}}{+1} + 2\frac{e^{-x}}{-1}\right]_0^{\infty}$

$1 = k\left\{\left[\cancel{x}0 - 0 + 0\right] - \left[0 \cancel{+} - 0 + 2\frac{e^{0}}{-1}\right]\right.$

$1 = k\left[2/-1\right]$

$1 = -2k \Rightarrow k = {}^{-1}\!/_2.$

$\text{Mean} = \int_0^{\infty} x f(x)$

$= k\int_0^{\infty} x\cdot x^2 e^{-x}$

$= k\int_0^{\infty} x^3 e^{-x}.$

$= k\left[x^3\frac{e^{-x}}{-1} - 3x^2\frac{e^{-x}}{+1} + \cancel{8}\,6x\frac{e^{-x}}{-1} \cancel{+} 6\frac{e^{-x}}{+1}\right]_0^{\infty}$

$= k\left\{\left[0 - 0 + 0 + 0\right] - \left[0 - 0 \cancel{+0} - 6\cdot e^{0}\right]\right.$

$= -6k \cdot -6\left(-1/2\right.$

$= +6/2$

$= +3.$

Q.No.

(ii) variance.

$$K\int_0^{\infty} x^2 \cdot x^2 e^{-x} - \mu^2$$

$$K\int_0^{\infty} \cancel{x^2 \cdot x}\, x^4 e^{-x} - (3)^2$$

$$K\int_0^{\infty}\left[x^4\frac{e^{-x}}{-1} - 4x^3\frac{e^{-x}}{+1} + 12x^2\frac{e^{-x}}{-1} - 24x\frac{e^{-x}}{+1} + 24\frac{e^{-x}}{-1}\right.$$

$$K\left[\left[0-0+0+24(0)\right) - \left[0+\frac{24}{-1}\right)\right]$$

$$= K[-24]$$

$$= \left(\frac{-1}{2}\right)(-24)$$

$$= 24/2$$

$$= 12$$

2b) $\int^1$ $\int_{\infty}^{0} f(x)dx + \int_0^1 f(x)dx + \int_1^2 f(x)\,dx + \int_2^3 f(x)dx) + \int_3^{\infty} f(x)dx$

$$0 + 0 + \int_1^3 k\,(x-3)^3\,dx.$$

$$K\int_1^3 (x-3)^3$$

$$K\int_1^3 x^3 + (3)^3 - 3a(1)($$

Q.No.

unit -1

1(a) first bag 1 white 2 red 3 green.

$p(A) = 2/3 \quad p(B) = 2/3 \quad , p(g) = 2/3.$

$p(B/A) = 2/3 \quad , p(B/B) = 2/5 \quad , p(B/g) = 2/0.4.$

Now

Baye's theorem

$$= \frac{p(AE_n) \cdot p(E_n/A_n)}{p(E_n) \cdot p(E_n/A) \cdot p(E_n) \cdot p(E_n/E_2) + p(E_n) p(E_n/E_3)}.$$

$$p(\text{second bag}/\text{Red and whit}) = \frac{p(W) \cdot p(B/W) + p(R) \cdot p(B/R)}{p(W)\, p(B/W) + p(R) \cdot p(B/R) + p(g)\, (p(B/g)}.$$

$$= \frac{(2/3) \cdot (2/3) + (2/3) \cdot (2/5)}{(2/3) \cdot (2/3) + (2/3)(2/5) + (2/3)(2/4)}.$$

$$= \frac{(0.6)(0.6) + (0.6) \cdot 0.4}{(0.6)(0.6) + (0.6) \cdot (0.4) + (0.6)(0.5)}.$$

$$= \frac{0.36 + 0.24}{0.36 + 0.24 + 0.3}$$

$$= \frac{0.6}{0.9} = 0.66.$$

Q.No.
Unit V

10 b) $3x + 12y = 19$

$3y + 9x = 46$

$x = \frac{-12y + 19}{3}$

~~3y =~~ $bxy = \frac{-12}{3}$

$3y + 9x = 46$

$y = \frac{-9x + 46}{3}$.

$byx = \frac{-9}{3}$.

$= \sqrt{\frac{-12}{3} \times \frac{-9}{3}}$.

$= \sqrt{4 \cdot 3}$

$= \sqrt{12}$

$= 3.46$

not valid.

$x = \frac{-3y + 46}{9}$

$= \frac{-3}{9}$

$= -3.$

~~$3y + 9(-3)$~~

$3y + 9\left(\frac{-3y + 46}{9}\right) - 46 = 0$

Q.No.

$27 - 27y + 414 - 444$

$3\left(\frac{-3y+46}{9}\right) + 12y - 19 = 0.$

$-9y + 138y + 108 - 171 = 0.$

$129y - 63 = 0.$

$129y = 63$

$y = 63/129$

$y = 0.48.$

$3(0.48) + 12y = 19.$

$= 1.44 + 12y = 19.$

$12y = 19 - 1.44$

$12y = 17.56$

$y = 17.56/12$

$= 1.46.$

$\sqrt{\frac{3}{12} \cdot \frac{9}{3}}$

$= \sqrt{0.25 \cdot 3}$

$= 0.866.$

valid.

Q.No.

Q.No.

Q.No.

Q.No.

Q.No.

Q.No.

Q.No.

Q.No.

Q.No.